What is Edge Computing

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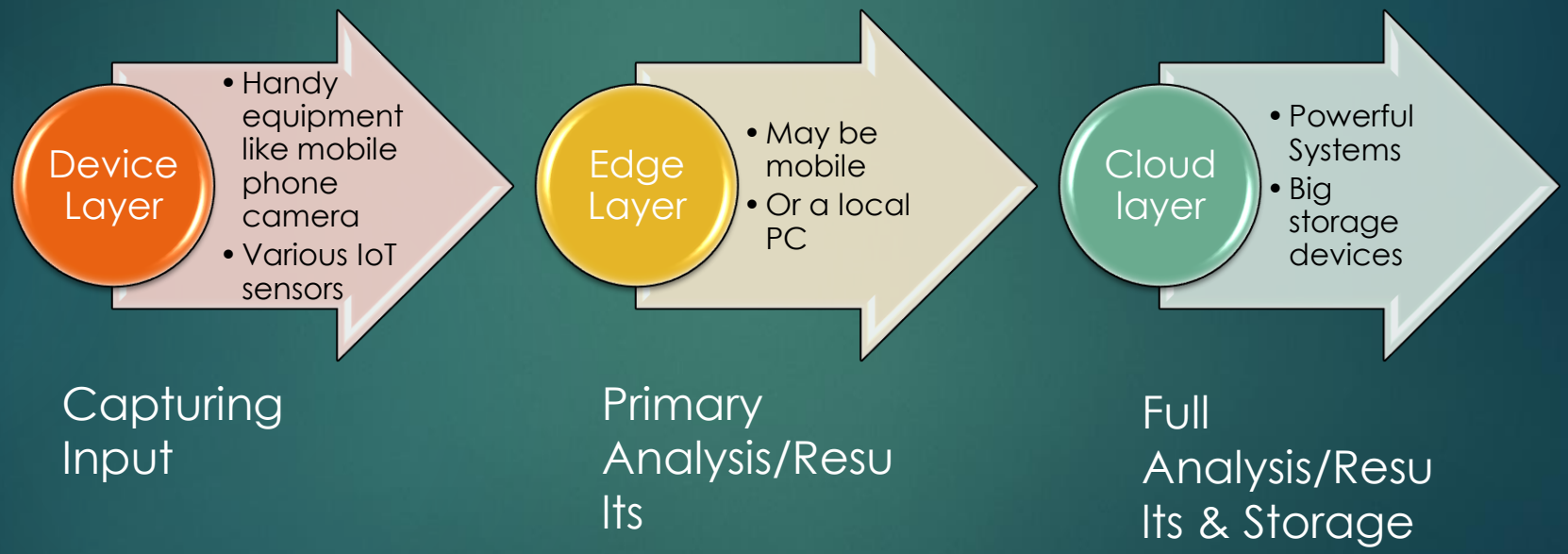
Limitation of Traditional Cloud Based Computing

Traditional cloud computing, all data is sent to a central cloud server. This creates issues like **high latency, dependency on internet, and delays**, especially in remote areas

Why Edge Computing?

- Processing near device
- Low latency
- Works with weak internet

Edge Architecture



Real-World Scenario (India)

Scenario : Former trying to identify his crop conditions/ crop planning

- Crops are regionally affected by surrounding disease
- They are affected by rapidly changing weather conditions and also Climatic condition
- May be there are some near by IOT sensors for soil conditions, sensors to monitor local environment – Limited Devices
- They are not very rich in terms of sensors or technical things usage also internet connection is not reliable with limited bandwidth – Poor Internet

Edge Solution

Device Layer

- Capture crop images
- Sensor Data Collection
- User Interaction

Edge Layer

- Sensor Data Reception
- Quick Analysis
- Instant Basic recommendations

Cloud Layer

- Stores large-scale data
- Performs long-term insights & predictions
- rains and updates models

One More term…… Fog Computing

- This is based on distributed computing.
- Data is processed at an intermediate layer between edge and cloud layer
- So , this is → processing data near the cloud not the device

Lets see this concept in our use case…

Edge + FOG Computing Solution

Device Layer

- Capture crop images
- Sensor Data Collection
- User Interaction

Edge Layer

- Sensor Data Reception
- Quick Analysis
- Instant Basic recommendations

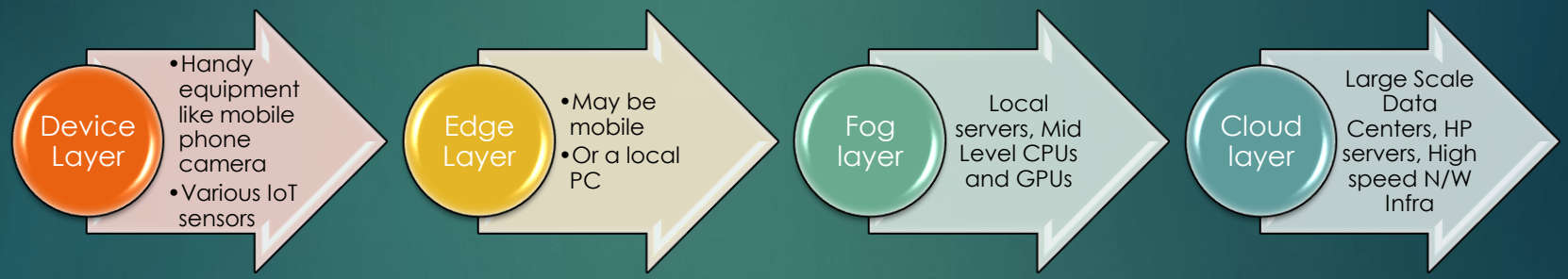
Fog Layer

- Regional Data Aggregation
- Regional Data Pattern Identification
- Send refined data to cloud for further processing

Cloud Layer

- Stores large-scale data
- Performs long-term insights & predictions
- rains and updates models

Edge + FOG Architecture



Capturing Input

Primary Analysis/Results

Regional Analysis & Intelligence

Full Analysis/Results & Storage

Conclusion (Edge + Fog)

- Enables **fast and real-time decision making** (Edge)
- Provides **regional intelligence and pattern detection** (Fog)
- Works efficiently with **low or unreliable internet**
- Improves **crop monitoring and disease detection**
- Supports **better planning using local + regional data**
- Reduces dependency on cloud → **faster response & lower bandwidth**

Thanks